Environment and Sustainability

III Semester SEE Question Papers

Course codes: 22CV232TA, CV232TA, CV242TA

- Sep/Oct 2024 (CV242TA)
- Apr/May 2024 (CV232TA)
- Model Question Paper (2022 Scheme) (22CV232TA)

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

IV Semester B.E. Examinations Sept./Oct. – 2024

Civil Engineering

ENVIRONMENT AND SUSTAINABILITY (BASKET COURSE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

PART-A

			M	BT	CO
1	1.1	List the components of environment.	02	1	1
	1.2	Differentiate between primary and secondary succession.	02	4	1
	1.3	Define climate change.	02	1	2
	1.4	List any two renewable and non renewable sources of energy.	02	1	2
	1.5	Define GDP as an economic indicator.	02	1	3
	1.6	Enlist any four Millennium Development Goal (MDG's).	02	1	3
	1.7	List any four ozone depleting substances.	02	1	2
	1.8	Define carbon foot print for a product or process.	02	1	3
	1.9	Write the principles of CSR.	02	2	4
	1.10	What are the two examples of business ethics?	02	2	4

PART-B

2	a	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	08	2	1
	b	Briefly describe the various Environmental protection acts promulgated in India.	08	2	2
3	a	Demonstrate on "Hydrogen energy is a clean energy", write the advantages and disadvantages of Hydrogen energy as an energy source.	08	2	2
	b	Enumerate the process of energy management and energy conservation with its objectives.	08	2	2
		OR			
4	a	Explain: i. Biomass energy ii. Ocean thermal energy conversion	08	2	2
	b	Define Energy cycle and explain with a neat sketch carbon cycle.	08	2	2
5	a	Discuss Environmental Economics and Environmental audit as a measure to sustainability.	08	2	3
	b	Discuss linear resource management systems as bane in sustainable Growth of industries.	08	2	3
		OR			
6	a	Illustrate how sustainable development is concept of three aspects of society.	08	2	2
	b	Define industrial ecology and explain the strategies to implement the same.	08	2	2

7	a	Enumerate all the 17 SDG's stipulated by United nations and brief the significance of any one.	08	1	3
	b	Explain Zero waste concept as a philosophical approach.	08	2	3
OR					
8	a	Describe Circular Economy and distinguish linear vs circular economy.	08	2	3
	b	Explain ISO 14000 and carbon Sequestration.	08	2	3
9	a	Recall and explain the history of evolution of CSR.	08	2	4
	b	Illustrate the concept of charity and corporate philanthropy.	08	2	4
OR					
10	a	Discuss the concept of sustainability and stakeholder Management.	08	2	4
	b	Enumerate the key features of Global Reporting Index and Dow Jones sustainability index.	08	1	4

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B.E. Examinations April/May – 2024

Common to All

ENVIRONMENT AND SUSTAINABILITY

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Define the terms Biotic and Abiotic components.	02	1	1
	1.2	Mention threats to bio-diversity.	02	1	1
	1.3	List out the types of land pollution.	02	1	2
	1.4	State the effects of noise pollution.	02	1	2
	1.5	What are fossil fuels and why are non-renewable?	02	1	2
	1.6	What is biogas and mention its uses.	02	1	2
	1.7	Highlight the significance of GDP in nation's economic activity.	02	2	3
	1.8	Highlight the distinction between Kyoto protocol and Paris agreement.	02	2	3
	1.9	Mention the principles of organic farming.	02	1	3
	1.10	Enumerate the significance of EIA in developmental projects.	02	1	4

PART-B

2	a	Discuss the importance of any four environmental protection acts promulgated in India.	08	2	1
	b	Briefly discuss the causes, effects and preventive measures for soil and air pollution.	08	2	2
3	a	Explain the concept of energy management and conservation.	08	2	2
	b	Enumerate the concept of sustainable urbanization and the need for the same.	08	1,2	3
		OR			
4	a	Explain the concept of hydrogen energy with its uses, advantages and disadvantages.	08	2	2
	b	Explain the carbon cycle with sketch.	08	2	2
5	a	Define environmental audit. Briefly describe the types of environmental audit.	08	1	3
	b	Enumerate the economic and social aspects of sustainability.	08	1	3
		OR			
6	a	Explain the key concepts of system thinking along with its advantages.	08	2	3
	b	Discuss the concepts of circular economy and industrial ecology.	08	2	3
7	a	Enumerate the various ways of reducing carbon footprint in our daily life.	08	1	3
	b	State and explain how green building concept evolved as sustainable habitat.	08	2	3
		OR			
8	a	Enumerate ISO 14000 series of standards.	08	1	3

	b	Explain in brief the necessity of energy efficiency and sustainable transport.	08	2	3
9	a	Explain the principles of CSR.	08	2	4
	b	Explain the concept of charity and corporate philanthropy.	08	2	4
OR					
10	a	Enumerate the environmental aspects of CSR.	08	1	4
	b	Explain how the changing emphasis of companies focusing on CSR.	08	2	4

22CV232TA

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B.E. Examinations MARCH – 2024

Model Question Paper

ENVIRONMENT AND SUSTAINABILITY (BASKET COURSE)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	Define the terms Eco-system and Bio-diversity.	02	–	–
	1.2	List various Energy cycles in nature.	02	–	–
	1.3	Illustrate what is carbon sequestration.	02	–	–
	1.4	Mention the significance to stop man wildlife conflicts.	02	–	–
	1.5	Describe the importance of GDP.	02	–	–
	1.6	Highlight any four threats to Bio-diversity.	02	–	–
	1.7	Mention any two major causes for climate changes.	02	–	–
	1.8	Suggest the significance of EIA as a tool for sustainable practices.	02	–	–
	1.9	What is performance measurement in business.	02	–	–
	1.10	Enumerate the prominence of CSR.	02	–	–

PART-B

2	a	Classify different types of ecosystems. Explain in brief any two ecosystems with their structure and composition.	08	–	–
	b	Describe the idealization made in Engineering Mechanics and also the characteristics of a force.	08	–	–
3	a	Explain carbon cycle with sketch.	08	–	–
	b	Enumerate the concept of sustainable urbanization and the need for the same.	08	–	–
		OR			
4	a	Explain ocean thermal energy conversion with its advantages and limitations.	08	–	–
	b	Explain nitrogen cycle with sketch.	08	–	–
5	a	What is environmental audit? Explain the steps in environmental audit.	08	–	–
	b	Enumerate the economic and social aspects of sustainability.	08	–	–
		OR			
6	a	Explain the linear and cyclic resource management systems.	08	–	–
	b	What are the advantages and limitations of industrial ecology?	08	–	–
7	a	Define the following i) Climate change ii) Carbon credit iii) Carbon footprint iv) Environmental impact assessment.	08	–	–
	b	Explain the concept of 5R.	08	–	–
		OR			
8	a	Enumerate ISO 14000 series standards.	08	–	–

	b	Define green buildings, enumerate the green building materials and write the advantages of green buildings.	08	–	–
9	a	Explain the principles of CSR.	08	–	–
	b	Explain the concept of charity and corporate philanthropy.	08	–	–
		OR			
10	a	Enumerate the environmental aspect of CSR.	08	–	–
	b	Explain how CSR has evolved in India.	08	–	–

Material Science for Engineers

III Semester SEE Question Papers

Course codes: ME232TB

- Apr/May 2024 (ME232TB)

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B.E. Examinations April/May – 2024

Common to All Branches

MATERIAL SCIENCE FOR ENGINEERS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	Ceramic materials are generally having _____ type of bonds.	01	1	1
	1.2	When the electrical behaviour of the material are dominated by the electronic structure of pure metal, the material can be called as _____ semi-conductors.	01	1	1
	1.3	Give an example of natural composite.	01	1	1
	1.4	T_i is having _____ crystal structure at room temperature.	01	1	1
	1.5	For the application of extremely high temperatures, _____ type of basic materials are used.	01	1	1
	1.6	$Cu - Ni$ alloys are an example of _____ type of defect.	01	1	1
	1.7	Deformations by twins are commonly observed in _____ type of metals.	01	1	1
	1.8	Calculate the co-ordination number of BCC crystal systems.	01	1	1
	1.9	External surfaces are an example of _____ type of defects in crystal systems.	01	1	1
	1.10	Viscoelastic deformation is observed significantly in _____ basic material.	01	1	2
	1.11	Diamond pyramid indenter is used in _____ hardness testing.	01	1	2
	1.12	Addition of chromium (Cr) to steel enhances _____ property.	01	1	2
	1.13	Which property of dielectric materials is measured in materials testing?	01	1	2
	1.14	Photodiodes serves as a _____ in optical communication systems.	01	1	2
	1.15	Name the hardening process which introduces nitrogen into the surface layer of steel to produce nitrides at the surface layer of the component.	01	1	3
	1.16	_____ is a critical process in electronics manufacturing primarily used in surface-mount technology.	01	1	3
	1.17	Aggregated composites possess _____ strength compared to fiber reinforced composites.	01	1	3
	1.18	Sol-gel process is an example of _____ approach of synthesis of nanomaterials.	01	1	4
	1.19	Crystalline structures of materials are measured on the basis of lattice spacing and electron interaction with it, in _____ characterization technique.	01	1	4
	1.20	Name characterization technique for nanostructure in which atomic structures of the materials are observed using electrons passing through ultra-thin specimens.	01	1	4

PART-B

2	a	Describe the following bonds for solid materials with examples. (i) Ionic bonds (ii) Secondary bonds and (iii) Metallic bonds.	08	2	1
	b	Explain the following crystallographic features of a solid material. (i) FCC crystal structure with example (ii) Energy bands in insulator (iii) Edge dislocation in a metal.	08	2	1
3	a	Define and explain the following thermal properties of a material. (i) Thermo-electric effect (ii) Linear thermal expansion coefficient (iii) Thermal shock.	08	2	2
	b	Explain the principle of working of following materials. (i) Piezoelectric materials and (ii) Thermocouples.	08	3	2
		OR			
4	a	With a stress-strain curve, explain the following properties of a material. (i) Young's Modulus (ii) Ultimate Tensile Strength and (iii) Toughness.	08	3	2
	b	Define the following mechanical properties of a solid material. (i) Fracture toughness (ii) Hardness (iii) Fatigue limit.	08	2	2
5	a	Explain the properties and applications of stainless steel.	06	2	2
	b	Mention the dopant used for p-type and n-type semiconductors.	04	2	2
	c	Explain the properties and applications of ceramics.	06	2	2
		OR			
6	a	Explain the properties and applications of fiber reinforced composites.	08	2	2
	b	Classify forming processes of structural materials and explain any two forming processes.	08	2	2
7	a	Explain the steps involved in thermal oxidation method of semiconductor devices.	08	3	3
	b	Describe the cause, effect and prevention methods of any two defects of heat treatment process of metals.	08	2	3
		OR			
8	a	Differentiate between annealing and normalizing process of ferrous materials.	08	3	3
	b	Describe hardening of steel with the help of TTT diagram.	08	2	3
9	a	Define nanomaterials and describe the applications of nanomaterials with example.	08	2	4
	b	Describe the effect of particle size and surface area of nanomaterials on the mechanical and optical properties of it.	08	2	4
		OR			
10	a	Describe the significant properties and applications of carbon nanotubes.	08	2	4
	b	Describe the principle of working of X-Ray Diffraction (XRD) techniques for the characterization of solid materials.	08	3	4

Biosafety Standards and Ethics

III Semester SEE Question Papers

Course codes: BT232AT, BT232TC

- Apr/May 2024 (BT232TC)
- Model Question Paper (2022 Scheme) (BT232AT)

BT232TC

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

V Semester B.E. Examinations April/May – 2024

Biotechnology

BIOSAFETY STANDARDS AND ETHICS

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	**M**	**BT**	**CO**
1	1.1	List the advantages of preserving of food with examples.	02	1	1
	1.2	Mention the dangerous temperature of food preservation.	02	1	1
	1.3	Mention the examples of food borne diseases.	02	1	3
	1.4	Give the importance of HEPA filters and examples.	02	1	4
	1.5	Write the role of antibiotics in Poultry.	02	1	1
	1.6	What is GHP? Explain its importance.	02	1	3
	1.7	What is food sterilization? Give examples.	02	1	4
	1.8	What is biosafety committee and its functions?	02	2	1
	1.9	List out the unit operations applicable in food processing with suitable examples.	02	1	1
	1.10	Differentiate between LAF and Biosafety Cabinet.	02	2	2

		PART-B			
2	a	Define biohazard and mention the various biohazards with example.	08	2	1
	b	Explain various biosafety levels using a triangular diagram.	08	2	2
3	a	Mention various functions of IBSC with examples.	08	3	2
	b	Define GMO. Explain 1st Generation, 2nd Generation GM food.	08	2	1
		OR			
4	a	Write short notes on i) RCGM ii) GEAC.	08	1	3
	b	Discuss on the salient features of Cartagena Protocol.	08	1	2
5	a	List out the important food safety Standards and Regulations.	08	2	2
	b	Mention the possible pathogens that can contaminate food and how do you prevent food contamination.	08	1	1
		OR			
6	a	Explain 7 basic principles of Hazard Analysis Critical Control Point (HACCP).	08	1	4
	b	Explain various biochemical analysis of food.	08	2	4
7		Explain chemical and physical method of food processing.	16	2	2
		OR			
8	a	Classify food packaging methods and write an explanatory note on the novel food packaging methods.	08	4	4
	b	Write short notes on GMP and GLP.	08	1	3
9	a	Discuss in detail about food and Drug administration.	08	5	2

	b	Write explanatory notes on food production and economics. Why is food production important to the economy?	08	2	3
OR					
10	a	Differentiate between clinical ethics and bioethics.	08	3	3
	b	Discuss the limits of additives to be added to the food as per the standards.	08	4	4

BT232AT

USN | | | | | | | | | | |

RV COLLEGE OF ENGINEERING®

(An Autonomous Institution affiliated to VTU)

III Semester B.E. April – 2024 Examinations

DEPARTMENT OF BIOTECHNOLOGY

BIO SAFETY STANDARDS AND ETHICS (2022 SCHEME) – MODEL QUESTION PAPER

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

		PART-A	M	BT	CO
1	1.1	What is level 2 biosafety cabinet?	01	–	–
	1.2	What are the allergens in the food? Give examples.	01	–	–
	1.3	Expand the term “BAP”.	01	–	–
	1.4	Explain the term “LMO”.	01	–	–
	1.5	Discuss the advantages of preservation of food.	01	–	–
	1.6	Cholera is caused by _____ microorganism.	01	–	–
	1.7	What is the most dangerous temperature of food preservation?	01	–	–
	1.8	What is the Food borne disease? Give two examples.	01	–	–
	1.9	Give the importance of HEPA filters.	01	–	–
	1.10	What is the role of antibiotics in Poultry?	01	–	–
	1.11	What is GHP? Explain its importance.	02	–	–
	1.12	How do you sterilize Food? Give suitable examples.	02	–	–
	1.13	What is biosafety committee and its functions?	02	–	–
	1.14	List out the unit operations applicable in food processing.	02	–	–
	1.15	Differentiate between LAF and Biosafety Cabinet.	02	–	–

PART-B

		UNIT-I			
2	a	Illustrate on biohazards and its consequences.	08	–	–
	b	Describe in detail various Biosafety levels, with the help of a neat pictorial representation.	08	–	–
		UNIT-II			
3	a	Elaborate on the institutional Biosafety norms.	08	–	–
	b	Classify GMOs and its applications in food and agriculture.	08	–	–
		OR			
4	a	Expand and give its functions i) RCGM ii) GEAC.	04+04	–	–
	b	Explain and discuss the features of Cartagena Protocol.	08	–	–
		UNIT-III			
5	a	List out Functions, types of Licenses and compliance rules of FSSAI.	08	–	–
	b	What is Food Hygiene? What are the possible pathogens that can contaminate food and how do you prevent food contamination?	08	–	–
		OR			

6	a	Illustrate the principles of Hazard Analysis Critical Control Point (HACCP).	08	–	–
	b	Give an account of Food Analysis and Testing in laboratories for maintaining the food quality.	08	–	–
UNIT-IV					
7	a	Discuss in details the need and importance of Food processing operations.	08	–	–
	b	Explain with an examples the various methods for food preservations.	08	–	–
OR					
8	a	Classify food packaging methods, and add and explanatory note on the novel food packaging methods.	08	–	–
	b	Describe Good Manufacturing Practices applicable to food and the advantages of GAP and GLP.	08	–	–
UNIT-V					
9	a	Discuss in detail about Food safety and ethics.	08	–	–
	b	Write explanatory notes on Food production and economics.	08	–	–
OR					
10	a	Write explanatory notes on Bioethics and Clinical Ethics.	08	–	–
	b	Classify food additives? Discuss the limits of additives to be added to the food as per the standards.	08	–	–